

Sewon Kim

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Seoul, South Korea

OBJECTIVE

Optimal Control and Robot Learning; Sophisticated Theories into the Real World.

EDUCATION

- **University of Seoul** Mar 01 2022 – Present
B.E. in Electrical and Computer Engineering, Minor in Mathematics Seoul, South Korea

EXPERIENCE

- **Institute of Innovation for Future Army, ROKA** Jan 13 2025 – Jul 12 2026 (Expected)
Drone Researcher (Compulsory Military Service) Daejeon, South Korea
- **InBody AI Scale, InBody** Jan 05 2024 – Jan 12 2025
AI Lead Engineer Seoul, South Korea
- **Control and Dynamic System Lab, UOS** Jan 31 2023 – Dec 23 2024
Fifteenth Lead Researcher; under the supervision of Prof. Gyunghoon Park Seoul, South Korea

PROJECTS

- **Autonomous air purifying robot agent** Aug 25 2025 – Sep 25 2026
4th Place (200 teams) — SK NAMUH X AI: Autonomous Driving Path Optimization (\$2,200) Seoul, South Korea
Independently implemented full-stack navigation purely in NumPy, entirely without open-source code, covering MCL, MPPI, and TSP.
- **Autonomous racing "Fifteenth"** Mar 3 2024 – Dec 19 2024
5th Place (12 teams) — 22nd Fifteenth Autonomous Grand Prix at CDC 2024 Milan, Italy
Led the project with 2 undergraduates and implemented a full-stack physical autonomous navigation system for racing completely from scratch, covering SLAM, MCL, Race Opt, MPPI.
- **Cart pole system design and control by PID and NMPC** Jul 25 2024 – Sep 27 2024
3rd Place (25 teams) — UOS ECE Innovation Fair (\$220) Seoul, South Korea
Independently developed a physical Cart-Pole system for \$272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.
- **New fully-automated body composition analyzer "InBody AI Scale"** Jan 1 2024 – Jan 11 2025
Consumer Electronic Show 2025 Las Vegas, USA
Developed height estimation model (accuracy 99.4%) and face recognition system of "InBody AI Scale".
- **ECG-based cardiovascular diagnosis device using ML** Jul 1 2023 – Jul 31 2023
3rd Place (12 teams) — InBody Hackathon 2023 (\$1450) Seoul, South Korea
- **Autonomous mobile robot for executing missions** Jul 24 2023 – Jul 28 2023
3rd Place (20 teams) — AI Object Detection-Based Solar Tracking Robot Challenge (\$220) Seoul, South Korea
- **Dating app "Sidaeting"** Mar 7 2022 – Jun 31 2022
1,700 User Base Seoul, South Korea
Developed the app's frontend of "Sidaeting," a dating service launched by UOSLIFE.

SKILLS

- **Technical Languages:** C, C++, Java, Julia, Python, JavaScript, SQL, HTML, CSS
- **Frameworks:** ROS1-2 (Noetic, Foxy, Humble), MuJoCo, PyTorch

LANGUAGES

- Korean
- English